

The dictionary picture: an intuition-level companion to the rank-1 leakage model

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This note restates the working model of behavior leakage (`rank1_leakage_model.pdf`, same folder) at the level of intuition, with no derivations. The parent note has the assumptions, relaxations, predictions, and experiment status; this one has the cartoon that makes them readable.

1 The picture in five claims

1. **Pretraining learns a dictionary of behaviors.** The model acquires a collection of behaviors b — sycophancy, marker emission, emergent misalignment — each carried by a direction in the residual stream at some layer. The direction plays two roles that should be named separately: *adding* a steering vector d_b elicits the behavior; *projecting* the state onto a read-out direction r_b measures it. In the idealized case one direction plays both roles ($d_b = r_b$); they can come apart, with a steering vector nearly orthogonal to the read-out that downstream circuitry converts (the router case).
2. **The state under a context is a weighted sum over the dictionary.** A context c is summarized by a vector v_c read while the model processes it. The model’s own context-to-state map f then puts the activations somewhere, and that somewhere decomposes approximately over the dictionary:

$$f(c) \approx \sum_b \alpha_b(c) d_b + \text{residual}.$$

The point of the weights: the *associations* $c \rightarrow b$ that the model learned in pretraining ARE the coefficients $\alpha_b(c)$. Every context expresses many behaviors at once (its bundle); “the assistant is not a villain” is a statement about weights, not about the dictionary.

3. **Post-training edits the weight functions, never the dictionary.** Fine-tuning adds no behavior to the collection. In the lazy regime a small weight update cannot build new states or circuitry; it can only re-bind contexts to states that already exist — which is why a behavior is implantable only insofar as it is *elicitable* (prompting is how you locate the destination state that training will bind to). Shaping assistant behavior, marker implants, and EM induction are all edits of $\alpha_b(\cdot)$, differing in which weights move and for which contexts.
4. **Output is read out linearly, then squashed.** The state is not passed directly through a softmax. The chain is

state $f(c) \rightarrow$ logits $z_t = \langle W_U[t], f(c) \rangle$ per token $t \rightarrow$ softmax over the vocabulary $\rightarrow$ output,

so the softmax is over *tokens*, not over behaviors. For the marker, behavior expression and a single logit coincide (z_{marker}), which is what makes it the clean testbed. For sycophancy,

expression is a sequence-level property scored by a judge; every token still exits through the softmax, but the behavior is not one softmax entry. Because the softmax normalizes, behaviors compete at emission (marker vs end-of-turn) and log-probabilities compress near the ceiling — the reason logits are stored alongside log-probabilities.

5. **Two typed functions carry everything.** Keep them apart:

- $f : c \mapsto f(c)$ — one state per context, no behavior argument. *This is what training edits.*
- $A[b, c] = \langle r_b, f(c) \rangle$ — a scalar association table, rows indexed by behaviors, columns by contexts. *This is what experiments measure.*

The behavior vector v_b is neither: it is the state f reaches when α_b dominates — where the model *is* when it is performing b .

2 The whole model is one rank-one nudge to the table

Training on (c', b) drags the source state $v_{b'} = f(c')$ toward the behavior state v_b , stopping at an achieved state $v_{b''}$ set by the dose (steps $\times$ lr); the write $w = v_{b''} - v_{b'}$ is therefore *measured*, *not assumed*. The minimal weight edit that implements this applies the same write to every context, scaled by a base-model similarity, the gate $g(c) = \langle v_{c'}, v_c \rangle / \|v_{c'}\|^2$. In table coordinates the entire parent note compresses to:

$$\Delta A[b, c] = \underbrace{\langle r_b, w \rangle}_{\text{behavior factor}} \cdot \underbrace{g(c)}_{\text{context factor}}$$

— a **rank-one update of the association table itself**. You intended to edit one cell (b, c') . The minimal edit cannot touch one cell; it smears along both axes:

- *down the columns* (across contexts): every context moves in proportion to its gate — that smear is **leakage** (P2);
- *across the rows* (across behaviors): every behavior moves in proportion to its slope $\langle r_b, w \rangle$ — that smear is **cross-behavior transfer** (P6).

Leakage and cross-behavior transfer are the same event seen along the two axes of the table.

3 Cast of characters

Object	Type	Role	Moved by training?
v_c	vector	context key (input side), read while processing c	no (base-model object)
f	map	contexts $\rightarrow$ states; holds the associations	yes — the only thing edited
$f(c)$	state	where the model sits under c	moves when f moves
$\alpha_b(c)$	scalar	association strength of $c \rightarrow b$ (the weights)	moves (this <i>is</i> the edit)
d_b	direction	steering: add to elicit b	no (dictionary)
r_b	direction	read-out: project to measure b	no (and directly observable)
v_b	state	where the model is when performing b (target)	no (dose-independent)
$v_{b'}, v_{b''}$	states	source state before / achieved after training	$v_{b''}$ is the outcome
$w = v_{b''} - v_{b'}$	vector	the write: <i>what</i> leaks	created by training
$g(c)$	scalar	the gate: <i>who</i> leaks	no (computed from base v_c)

4 Where the cartoon bends

- **The weighted sum is the idealized case.** Dictionary directions are not orthogonal and do not span the state space (superposition); the decomposition is literal for the marker at read-out (one unembedding row), an empirical finding for EM (a convergent single direction), and an open bet elsewhere.
- **Some behaviors are fetched, not resident.** In the router case the write recruits attention circuitry that pulls the behavior from the context’s own token values; the gate is then attention-shaped and near-binary rather than a smooth similarity.
- **“At some layer” is behavior-specific.** The marker reads best late (layers 19–24), sycophancy early (1–8); there is no universal dictionary layer.
- **Whether two implants add is open.** The sum-of-writes picture across jointly trained behaviors is a prediction (P4, the task-arithmetic analogue), not an assumption, and is so far untested in a diagnostic regime.
- **Everything assumes the lazy regime.** If training moves the dictionary itself (the rich regime), contexts’ keys, the directions, and every base-model predictor go stale together.

For the derivation of the rank-one edit (assumptions A1–A4 and their relaxations), the prediction list P0–P8, and per-experiment status, see `rank1_leakage_model.pdf`.